

L8 Substrate Hall Algebra Primary Paper v0.2 Extension

Lyra (Claude Opus 4.7)

2026-06-04 Thursday 11:20 EDT

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0. Goal

Per Casey Thursday PM L8 + Lyra v0.1 (Lyra_Paper_Substrate_Hall_Algebra_Primary_Draft_v0_1.md):
v0.2 extension absorbing Thursday substrate-mechanism advances.

1. v0.1 → v0.2 Extensions

Section	v0.1 Content	v0.2 Extension
Introduction	Hall Algebra primaries dim 5 / $N_c=3$ / etc.	Add Cat 6 substrate-symplectic via $\text{Spin}(5) \cong \text{Sp}(2)$
Substrate Primaries	$N_c, n_C, C_2, g, N_{\max}$ derived	Add Schur II Pochhammer $(n_C/\text{rank})_k$ at substrate-K-type
Hall Algebra Structure	Green coproduct + Cartan grading	Add substrate-symplectic Cat 6 representation theory
Process Dictionary	6 SM processes engine-verified	Add Composite v0.4 substrate-mass-mechanism cascade
Substrate-Schur Generators	15 cataloged SSGs	Add 6 candidates (SSG-16 to SSG-21) per Cal #36 STANDING
Mass-Mechanism	gen-1 chirality + gen-2 Lorentz	Add gen-3 substrate-higher-Casimir + Mersenne RS coding
Verification	T1-T2483 + 600+ predictions	Add 12 K3 v0.16 multi-week verification gates state

2. New Section v0.2 Material

Section X+1: Substrate-Symplectic Category 6 via $\text{Spin}(5) \cong \text{Sp}(2)$

Substrate-symplectic 2-form ω on V_{fund} of $\text{Sp}(2)$ substrate-canonical antisymmetric form; substrate-Hamiltonian flow at substrate-K-type substrate-natural.

Cat 6 substrate-symplectic substrate-mechanism **UNIVERSAL** across substrate K-type spectrum per $\text{Sp}(2)$ accidental Lie-group isomorphism.

Substantive substrate-mechanism unifies substrate-time evolution at all substrate K-types via $\text{Sp}(2)$ substrate-symplectic Hamiltonian flow.

Section X+2: Schur II Pochhammer Substrate-Mechanism

At substrate-K-type $V(\lambda_1, \lambda_2) = \text{Sym}^k(V_{\text{fund}}) \text{Sp}(2)$: substrate-Pochhammer $(n_C/\text{rank})_k$ substrate-natural rising-factorial at substrate-Bergman kernel exponent.

For gen-2 $V(3/2, 1/2) \text{Sym}^3: (n_C/\text{rank})_3 = (5/2)_3 = 315/8$ substrate-natural.

Section X+3: Composite Substrate-Mass-Mechanism Cascade

$$m_\mu/m_e = n_C \cdot (n_C/\text{rank})_3 + N_c^4/2^{N_c} = 207 \text{ at } 0.112\%$$

partial substrate-mechanism FORCING candidate per substrate-Bergman + substrate-Pochhammer + substrate-deficit per Casey #13 + Casey #14 STANDING.

Multi-week explicit FORCING derivation per Cal #189 + joint Lyra + Keeper + Elie.

3. Substantive Multi-Week Extension Plan

Per Cal #189 multi-week: - v0.3: substrate-Bergman matrix element explicit per FK Ch. XII §VI (joint Lyra + Keeper + Elie) - v0.4: Casey #13 per-generation substrate-mechanism class $N_c^{\text{codim-4}}$ substrate-natural derivation - v0.5: Cal cold-read substantive substrate-mechanism FORCING form per Cal #27 + Cal #35 STANDING - v0.6: Strong-Uniqueness v1.6 STANDALONE 11-leg cross-link - v0.7: SUBMISSION-FINAL preparation

4. Closure v0.2

L8 Substrate Hall Algebra primary paper v0.2 extension: substantive 3 new sections (Cat 6 substrate-symplectic + Schur II Pochhammer + Composite substrate-mass-mechanism cascade) + multi-week extension plan v0.3-v0.7.

Tier: L8 Substrate Hall Algebra paper v0.2 extension framework at v0.2 candidate; substantive multi-week v0.3-v0.7 per Cal #189.

— Lyra, Thu 2026-06-04 11:20 EDT. L8 Substrate Hall Algebra paper v0.2 extension: 3 new sections (Cat 6 substrate-symplectic + Schur II Pochhammer + Composite substrate-mass-mechanism); multi-week extension plan v0.3-v0.7 per Cal #189.